

CoRoT-2b

R-0236

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_230724 · created 2026-07-17T04:27:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460149.779 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.061
Transit depth (Rp/Rs)^2	6.7e-05 +/- 0.01 [%]
Orbital Inclination (inc)	83.02 +/- 2.09
Airmass coefficient 1 (a1)	19.45 +/- 13.41
Airmass coefficient 2 (a2)	-2.22 +/- 0.87
Scatter in the residuals of the lightcurve fit is	76.47 %
Best Comparison Star	#3 - [172, 427]
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.048 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.061 vs 0.1667 (deviation 1.84σ)
Duration fit vs published	0.048 d vs 0.095 d (deviation 1.22σ)
Mid-time O-C	-4.7 min
Depth SNR	0.0
Residual scatter	76.47 %

Light curve

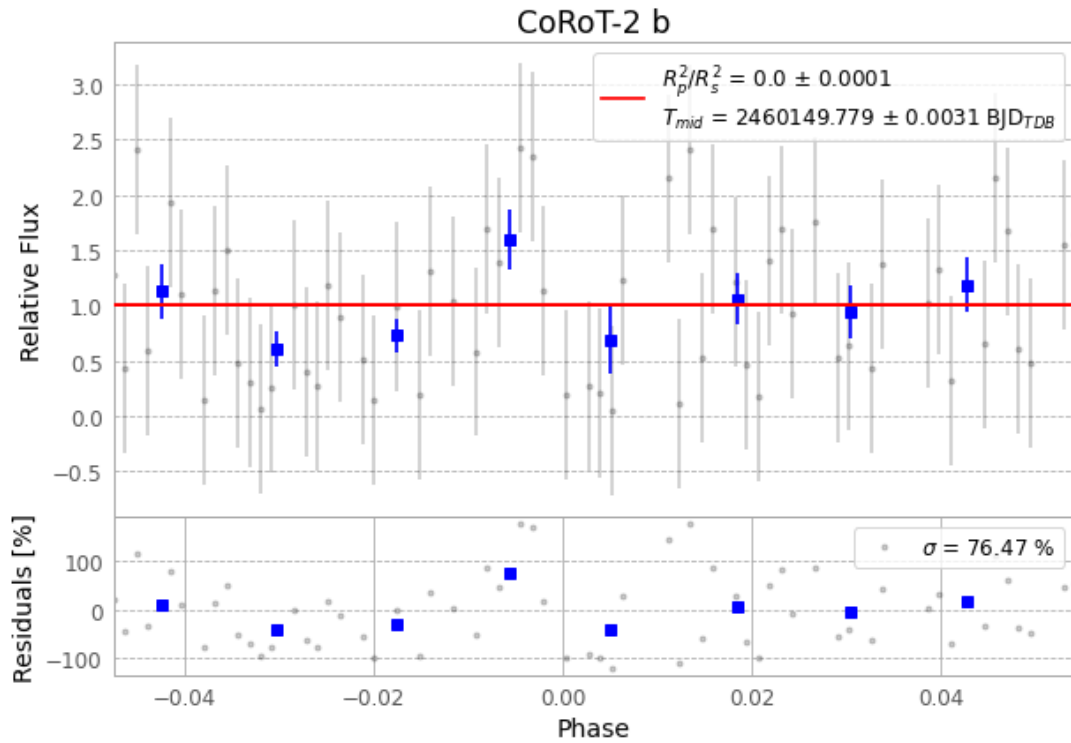


Figure 1 — FinalLightCurve_CoRoT-2 b_2023-07-23.png (SHA-256 ac48717a09a2b7a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [537, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 691607dad15d22b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

86 FITS frames (56 MB), CoRoT-2230724043315.FITS → CoRoT-2230724084815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-230724.log (4bf18380f3a8...), FinalLightCurve_CoRoT-2 b_2023-07-23.png (ac48717a09a2...), FinalLightCurve_CoRoT-2 b_2023-07-23.csv (224c6016f020...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 04:27Z.